

Zilin Xiao

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Education

Rice University, Houston, TX

2023 – 2026 (Expected)

Ph.D. student in Computer Science with Prof. Vicente Ordonez

Plan to graduate in December 2026 with the support of the advisor.

Sichuan University, Chengdu, China

2018 – 2022

B.Eng. in Computer Science

Experience

Research Scientist Intern

Menlo Park, CA

Meta Superintelligence Labs with Dr. Qi Ma and Dr. Vijai Mohan

May 2025 – Dec 2025

- Focus on scaling multi-modal embedding models at test-time.
- Related publications: test-time scaling in multi-modal retrieval at [arxiv'3]

Research Intern

Bellevue, WA

Tencent AI Lab Seattle with Dr. Hongming Zhang and Dr. Tao Ge

May 2024 – Aug 2024

- Focus on large language model inference efficiency in the agentic framework.
- Related publications: parallel speculative decoding at [arxiv'1], enhancing agent framework for software engineering at [ICLR'1]

Residency Research Intern

Beijing, China

Microsoft with Dr. Ming Gong

Aug 2022 – Jun 2023

- Focus on aligning large foundation models with entity-related information retrieval tasks.
- Related publications: predicting entity as tokens at [EMNLP'1], generative entity linking at [EMNLP'2]

Research Assistant

Houston, TX

VisLang Lab at Rice University

Sept 2023 – Now

- Focus on augmenting language models with visual recognition and reasoning abilities.
- Related publications: generative visual entity recognition at [ECCV'1], image retrieval with sequence modeling at [ECCVW'1], [CVPR'1], reinforcement learning with verifiable reward on visual reasoning at [arxiv'2]

Selected Publications

- [EMNLP'1] [Zilin Xiao](#), Linjun Shou, Xingyao Zhang, Jie Wu, Ming Gong, Jian Pei, Daxin Jiang. Coherent Entity Disambiguation via Modeling Topic and Categorical Dependency.
In the 2023 Findings of Empirical Methods in Natural Language Processing (Findings of EMNLP 2023)
- [EMNLP'2] [Zilin Xiao](#), Ming Gong, Jie Wu, Xingyao Zhang, Linjun Shou, Jian Pei, Daxin Jiang. Instructed Language Models with Retrievers Are Powerful Entity Linkers.
In the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)
- [ECCV'1] [Zilin Xiao](#), Ming Gong, Paola Cascante-Bonilla, Xingyao Zhang, Jie Wu, Vicente Ordonez. Grounding Language Models for Visual Entity Recognition.
In the 18th European Conference on Computer Vision (ECCV 2024)
- [COLM'1] [Pengda Wang](#)[†], [Zilin Xiao](#)[†], Hanjie Chen, Frederick L. Oswald. Will the Real Linda Please Stand up...to Large Language Models? Examining the Representativeness Heuristic in LLMs
In the 2024 Conference on Language Modeling (COLM 2024) (Oral Spotlight). [†]: equal contribution.
- [ECCVW'1] [Zilin Xiao](#), Ayush Sachdeva, Hao-Jen Wang, Vicente Ordonez. Image Re-ranking with Long-Context Sequence Modeling
In the Instance-Level Recognition Workshop at ECCV 2024 (Short Paper)
- [arxiv'1] [Zilin Xiao](#), Hongming Zhang, Tao Ge, Siru Ouyang, Vicente Ordonez, Dong Yu. ParallelSpec: Parallel Drafter for Efficient Speculative Decoding.
Preprint

7. [\[ICLR'1\]](#) [🔗](#) Siru Ouyang, Wenhao Yu, Kaixin Ma, **Zilin Xiao**, Zhihan Zhang, Mengzhao Jia, Jiawei Han, Hongming Zhang, Dong Yu. RepoGraph: Enhancing AI Software Engineering with Repository-level Code Graph.
In the 2025 International Conference on Learning Representations (ICLR 2025)
8. [\[CVPR'1\]](#) [🔗](#) **Zilin Xiao**, Pavel Suma, Ayush Sachdeva, Hao-Jen Wang, Giorgos Kordopatis-Zilos, Giorgos Tolias, Vicente Ordonez. LOCORE: Image Re-ranking with Long-Context Sequence Modeling
In the IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (CVPR 2025)
9. [\[arxiv'2\]](#) [🔗](#) **Zilin Xiao**, Jaywon Koo, Siru Ouyang, Jefferson Hernandez, Yu Meng, Vicente Ordonez. Proxy-Thinker: Test-Time Guidance through Small Visual Reasoners
Preprint
10. [\[NeurIPS'1\]](#) [🔗](#) Siru Ouyang, Xinyu Zhu, **Zilin Xiao**, Minhao Jiang, Yu Meng, Jiawei Han. RAST: Reasoning Activation in LLMs via Small-model Transfer
In the Advances in Neural Information Processing Systems 2025 (NeurIPS 2025).
11. [\[arxiv'3\]](#) [🔗](#) **Zilin Xiao**, Qi Ma, Mengting Gu, Jason Chen, Xintao Chen, Vicente Ordonez, Vijai Mohan. MetaEmbed: Scaling Multimodal Retrieval at Test-Time with Flexible Late Interactions
Preprint

Academic Service

Teaching Assistant: 2024 Spring – Deep Learning for Vision and Language [🔗](#)

- Guest Lecture: What could be the Next Generation of LLMs? [🔗](#)

Teaching Assistant: 2025 Spring – Reinforcement Learning

Reviewer / Program Committee: EMNLP 2023, ACL ARR, NeurIPS, ICLR, ICML, AISTATS, AAAI

Selected Awards

National Scholarship, Ministry of Education, P.R. China	<i>Oct 2019</i>
National Scholarship, Ministry of Education, P.R. China	<i>Oct 2020</i>
Outstanding Undergraduate Thesis Award, Sichuan University	<i>Jun 2022</i>
Star of Computer Science (Outstanding Graduate), Sichuan University	<i>Jun 2022</i>
2024 Travel Grant Award, Rice Engineering Alumni	<i>Oct 2024</i>